SMART INDIA HACKATHON 2026

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SkillSaathi

From Career Confusion to Family Consensus

- **Problem Statement ID – 26241**
- **Problem Statement Title –** AI-Enabled Career Counselling and Family Decision-Support Platform for Vocational Education
- **Theme –** Smart Education
- **PS Category –** Software
- **Team ID –** [To Be Added]
- **Team Name (Registered on portal) – TeamChargers**



TeamChargers

IDEA TITLE: SkillSaathi — *Beyond Career Advice. Towards Family Consensus.*

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An AI-powered family counselling platform that bridges student aspirations and parental concerns through evidence-backed vocational guidance.

PROPOSED SOLUTION: FAMILY DECISION INTELLIGENCE WORKFLOW

English + Regional Language | Simple, Family-Friendly Guidance

Student Aspirations
Interests, Aptitude & Ambitions

Parent Concerns
Financial Security, ROI & Stability

AI FAMILY COUNSELLING
Dual-Persona Dialogue Mediation
- Identifies stated concerns & priorities
- Synthesizes passion with practical ROI

CAREER COMPARISON
Evidence-Based Benchmarking
- NSQF-aligned qualifications & training
- Proposed placement & salary evidence

INFORMED FAMILY DECISION
Consensus & Career Clarity
- Aligned student-parent commitment
- Planned pre/post confidence tracking

INNOVATION & UNIQUENESS — WHY SKILLSAATHI?

01. Family Consensus Intelligence ★ FLAGSHIP USP
Identifies family disagreements, addresses parental objections, and tracks changes in family confidence.

Student | Parent

Identify Disagreements — Student vs Parent View → **Compare Trade-Offs** — Evidence Fact-Checking → **Shared Family Decision** — Pre/Post Confidence Track

02. Hyperlocal Career Reality Engine
Connects vocational choices to regional earnings, placement outcomes, and accredited training pathways.

District Cluster — Local ITI & Hubs → **₹ Regional Wages** — Estimated Wage Data | **Placement Trends** — Proposed Sources | **NSQF Pathway** — ITI → Poly → B.Voc

03. Explainable Career Trade-off Engine
Compares career options based on family priorities, training costs, earnings, and NSQF progression.

Family Priorities — Budget & Values | **Training Cost** — Fees vs Subsidy | **Training Duration** — Time-to-Earn Clarity | **NSQF Pathway** — Progression Options

04. Parental Resistance Intelligence
Transforms counselling concerns into aggregated insights and escalation pathways for scheme administrators.

Unresolved Hesitation — Stigma / Financial Anxiety → **Human Counsellor Routing** — 1-on-1 Escalation Hand-off + **Admin Heatmaps** — Participating Insights

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TECHNICAL APPROACH: System Architecture & Implementation

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Proposed end-to-end architecture for evidence-grounded family career counselling.

STAGE 01 • React Frontend [Active]

User Interface

- React + TypeScript + Vite [Active]
- Student & Parent Intake Interfaces
- Course Exploration & Rule Matching
- English, Hindi, Marathi Localization

STAGE 02 • FastAPI Services [Planned]

Backend & Data

- FastAPI Architecture [Planned]
- TypeScript Data Contracts Scaffold
- Persistent Family Profiles & Sessions
- Planned Integration of Curated Data

STAGE 03 • Core Innovation [Proposed]

AI Decision Engine

- Evidence-Grounded AI Counselling
- Student-Parent Disagreement Detection
- Explainable Career Trade-offs
- Pre/Post Confidence Assessment

STAGE 04 • Action & Governance [Proposed]

Outcomes & Analytics

- Joint Family Decision Summary
- Human Counsellor Escalation
- Participating Family Analytics
- AI Quality Evaluation Planned

Official Benchmarks ➔
(NCVET & NCS Data)

DATA GROUNDING ENGINE
Planned Integration of Curated Official Vocational Data

TECHNOLOGY STACK & IMPLEMENTATION SCOPE

FRONTEND [ACTIVE IN CODE]
React, TypeScript, Vite

Modular component library, student/parent intake forms, course explorer & trilingual UI.

BACKEND [PLANNED ARCHITECTURE]
FastAPI (Planned)

TypeScript data contracts & REST client scaffold implemented; FastAPI/Pydantic backend planned.

AI & DATA [PROPOSED LOGIC]
Grounded AI + Curated Data

Evidence-grounded counselling proposed using verified vocational standards; local baseline data in prototype.

OUTPUTS [PROPOSED SYSTEM]
Counselling & Analytics

Joint decision summaries, human escalation handoff, and participating family insights.

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FEASIBILITY AND VIABILITY

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Execution Roadmap & Risk Mitigation — Demonstrating practical implementation through a 3-phase progression, grounded data controls, and modular architecture.

SECTION 1: 3-PHASE IMPLEMENTATION ROADMAP *[From Live Prototype to Phased Field Deployment]*

01 PHASE 01 — FOUNDATION [ACTIVE IN CODE]

- **React + TypeScript Frontend:** Operational responsive UI, student & parent intake forms, trade explorer
- **Trilingual Localization:** English, Hindi & Marathi interfaces implemented in frontend
- **Course Exploration Engine:** Course exploration using structured demonstration data

02 PHASE 02 — INTELLIGENCE [PLANNED ARCHITECTURE]

- **FastAPI Backend Integration (Planned):** REST API client, session state persistence & data contracts
- **Evidence-Grounded AI Engine (Planned):** Proposed counselling anchored to official qualification standards
- **Family Disagreement & Confidence Tracking:** Planned consensus scoring & objection analysis

03 PHASE 03 — DEPLOYMENT [FUTURE PILOT ROLLOUT]

- **Rural Usability & Vernacular Testing (Future):** Field testing across diverse demographic cohorts
- **Human Counsellor Escalation (Future):** Automated triage routing complex family hesitation to experts
- **Admin Analytics & Pilot Testing (Future):** Participating family insights for scheme planners

SECTION 2: CHALLENGES & MITIGATION MATRIX *[Technical Safeguards & Risk Controls]*

Challenge	Mitigation
Limited Verified Data — Uncurated regional wage reports	**Curated Official-Source Ingestion** — Curated official-source ingestion; API access subject to availability.
AI Hallucinations — Unverified claims or false guarantees	**Grounded Retrieval & Evidence Checks** — Source citations, evidence checks, evaluation and human escalation.
Language Barriers — Vernacular hesitation among rural parents	**Regional UI & Simplified Terms** — English/Hindi/Marathi UI tested with plain, family-friendly terms.
Data Privacy Concerns — Sensitive family financial ambitions	**Consent Intake & Minimal Retention** — Planned access controls, consent, minimal retention and session testing.
System Scalability — Traffic surges during admission cycles	**Modular Backend Architecture** — Modular backend architecture + phased pilot rollout.

FEASIBILITY PRINCIPLE: Existing Frontend + Modular AI Architecture + Phased Deployment = Practical Implementation

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IMPACT AND BENEFITS

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Transforming Vocational Decision-Making — Replacing family hesitation with evidence-backed consensus, clear pathways, and institutional alignment.

EXPECTED IMPACT — TO BE VALIDATED IN PILOT

BEFORE SKILLSAATHI *[Current Hesitation Reality]*

- Career Confusion
- Parental Hesitation
- Uncertain Earnings
- Fragmented Guidance



SKILLSAATHI

→

AI Evidence Mediation

WITH SKILLSAATHI *[Evidence-Backed Consensus]*

- Clear Vocational Pathways
- Evidence-Backed Discussions
- Localized Career Insights
- Joint Family Counselling



SECTION 2: MULTI-STAKEHOLDER VALUE CREATION *[Human-Centered Benefits Across the Vocational Skilling Ecosystem]*

01 — STUDENTS
Career Clarity & Opportunity

- Personalized career clarity
- NSQF progression awareness

02 — PARENTS & FAMILIES
Trust & Informed Decisions

- Evidence-backed confidence
- Shared family decisions

03 — TRAINING INSTITUTIONS
Better-Informed Enrolment

- Better-informed applicants
- Potential retention improvement

04 — GOVERNMENT / MSDE
Data-Driven Skilling Decisions

- Regional resistance insights (participating families)
- Targeted counselling outreach

SECTION 3: PROPOSED IMPACT PATHWAY *[Sustainable Impact from Family Decision to Economic Growth]*

Family Confidence (STEP 1)
Joint dialogue helps families discuss concerns

→

Informed Enrolment (STEP 2)
Learners choose verified vocational tracks

→

Improved Retention (STEP 3)
Family backing supports course completion

→

Skilled Workforce (STEP 4)
Employable, certified youth for industry

" SkillSaathi empowers not just individual learners, but the families shaping their futures. "

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RESEARCH AND REFERENCES

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Foundational Policy & Academic Frameworks — Grounding SkillSaathi's vocational taxonomies, wage baselines, and family mediation models in verified standards.

SECTION 1: OFFICIAL FRAMEWORKS & RESEARCH REFERENCES *[IEEE Citation Format]*

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[2] **Ministry of Skill Development and Entrepreneurship, "Ministry of Skill Development and Entrepreneurship,"** *(National Vocational Policies & Scheme Mandates)*
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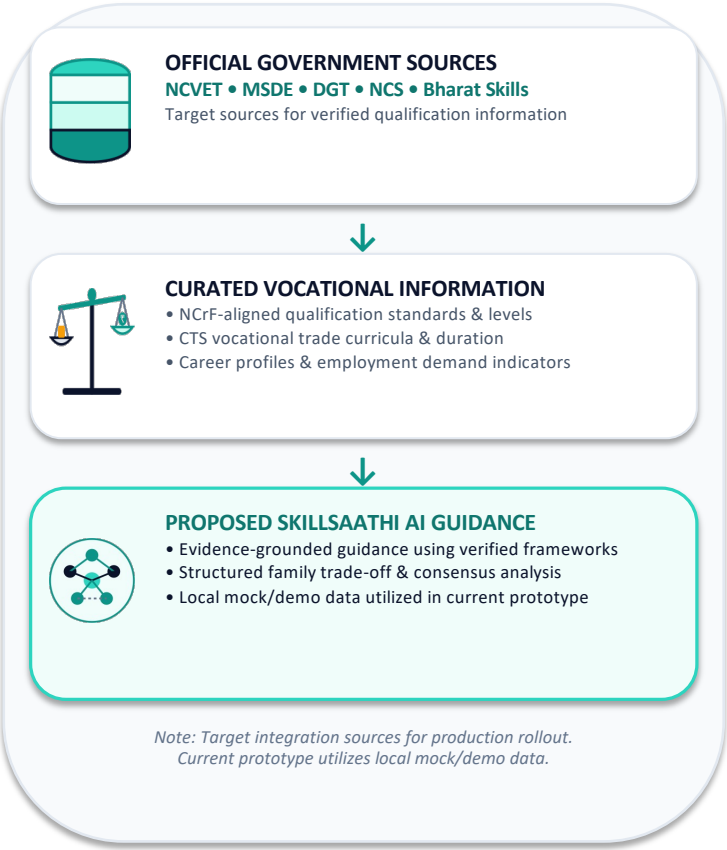
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[6] **J. Dietrich and B. Kracke, "Career-specific parental behaviors in adolescents' development,"** *(Parental Influence on Career Decisions — Academic Research)*
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SECTION 2: DATA GROUNDING ARCHITECTURE



" SkillSaathi prioritizes verified government frameworks and evidence-grounded recommendations to improve the reliability of vocational career guidance. "